

FACT-FILLED AND THE MOST HAPPENING CURRENT AFFAIRS FROM AROUND THE WORLD

Here comes spa robot

Get ready for head massage by a robot. Developed by Panasonic, the new robot will give you dry head spa treatment.

All you need to do is to adjust your head position and the hairline. The robot will then scan your head in order to develop a 3D map with the use of special sensors. This map helps robot's 24 fingers to apply varying and correct pressure at all the relevant pressure points on your head.

The robot can easily expand and contract its arms to reach all the relevant pressure points on your head.

Panasonic is now planning to couple this spa robot with the massage chair to provide users with in-chair spa treatment.

Open Source robot

Professor Sven Behnke and his team from the University of Bonn have come up with an Open Source robot named NimbRo-OP. Professor Behnke's team had previously won the Louis Vuitton Cup for 'Best Humanoid' in RoboCup 2012. The new robot is based on the same NimbRo model but has an attractive white head.

NimbRo-OP is 95cm tall and weighs 6.6 kg. It is powered by a dual-core AMD E-450 processor along with 2 GB of RAM and a 64GB solidstate drive. The robot is equipped with WiFi, a wide-angled video camera (Logitech C905) and a Robotis CM-730 controller with 3-axis accelerometer and 3-axis gyroscope.

The software is released under free and open licences. The GNU/Linux software is based on DARwin-OP. As this robot project is Open Source, other groups can further improve it and carry to upcoming RoboCup events. The next RoboCup will be held in June 2013 in the Netherlands.



Panasonic's prototype dry head spa robot, which in the future could be attached to a desk or bath. It uses robot hand technology previously developed for use in their hair washing robot. (Courtesy: www.houseofjapan.com)

Tennis player robot

Researchers at the Technical University of Darmstadt in Germany have come up with a robot that not only learns to play ping-pong from humans but also shows improvement as it competes against them. The robot is said to have all the potential to become the best robotic table-tennis challenger the world has ever seen.

The researchers suspended the robotic arm from the ceiling and equipped it with a camera that watched the playing area. The arm was physically guided through different shots to return incoming balls, then left to draw on its training to return balls hit by a human opponent. When the ball was in a position it had not seen before, the arm used its library of shots to improvise new ones.

After an hour of practice, the test arm successfully bounced back 88 per cent of shots.

New glove lets your hand function as a wireless keyboard

Researchers from the University of Alabama have come up with a smart glove device that turns your hand into a wireless keyboard.

When using typical keyboards, users need to tap keys on them, which

is generally not easy one-handedly or while walking. But with Gauntlet, or the 'generally accessible universal nomadic tactile low-power electronic typist,' the user can simply touch his thumb to certain points on his fingers that are assigned some or the other keyboard function.

Conductive thread carries the commands to a matchbox-sized printed circuit board (PCB) affixed to the back of the glove. The PCB transmits it via Bluetooth, to a computer, mobile phone, music synthesiser, video game or military device.

"It's basically a keyboard on your hand. You, by tapping your thumb on each segment of your fingers, type to the screen basically. And you can do a swiping gesture that would erase it," research team leader Jiakui Lui said. "There are several applications we can think of right now. The easy one would be a keyboard for the consumer market," he added.

Technology that turns paper into digital display

At University of Tokyo, the Naemura Group is developing a paper computing technology that lets users draw, edit and erase sketches on paper